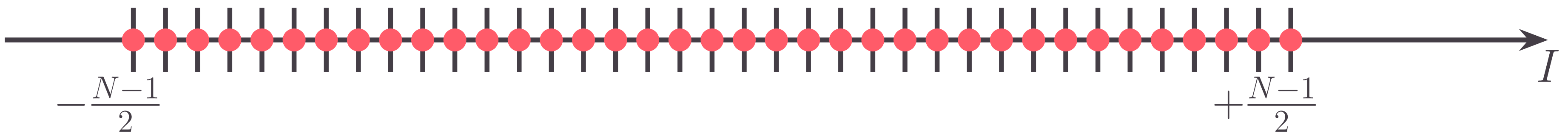
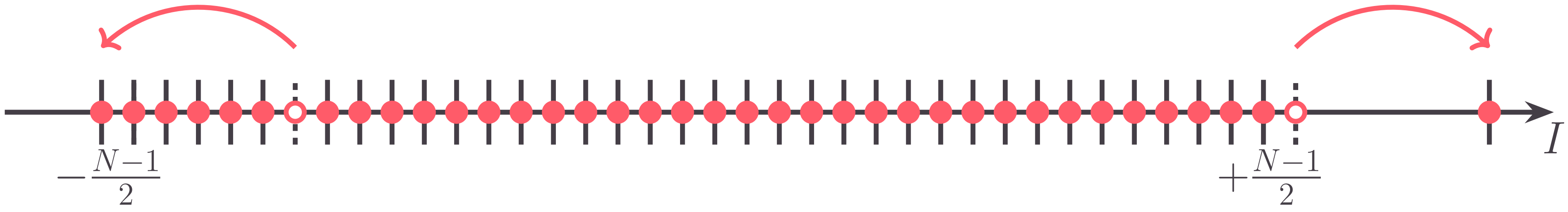
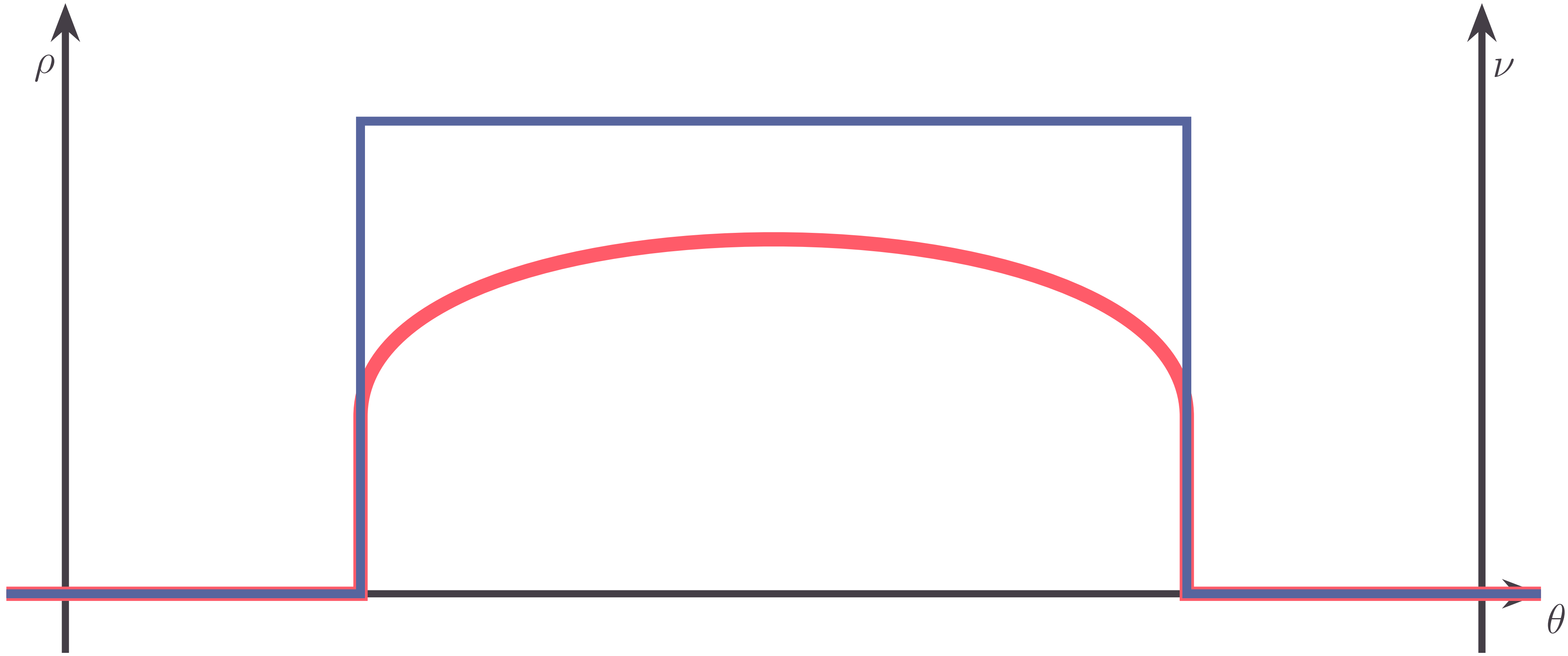
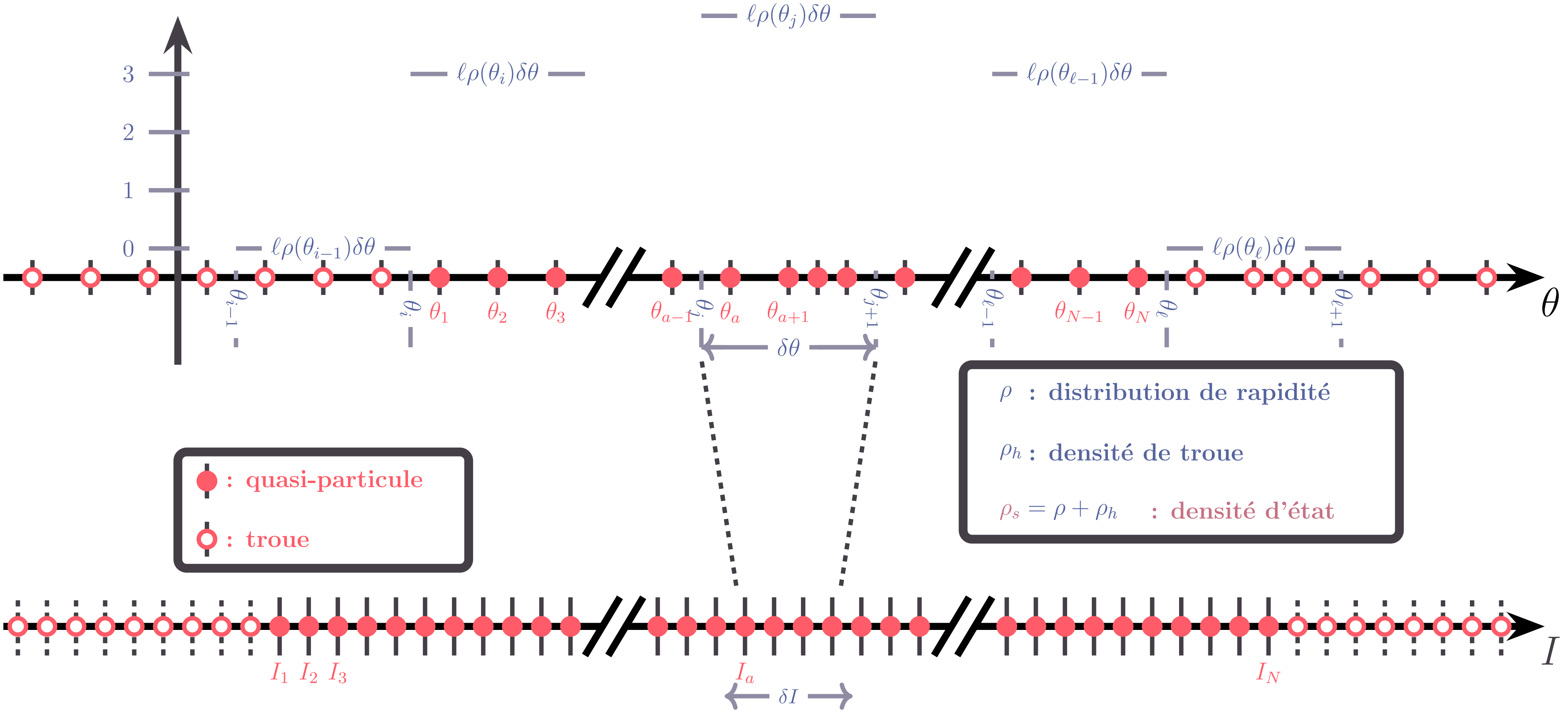

$-\frac{N-1}{2}$
$+\frac{N-1}{2}$
I



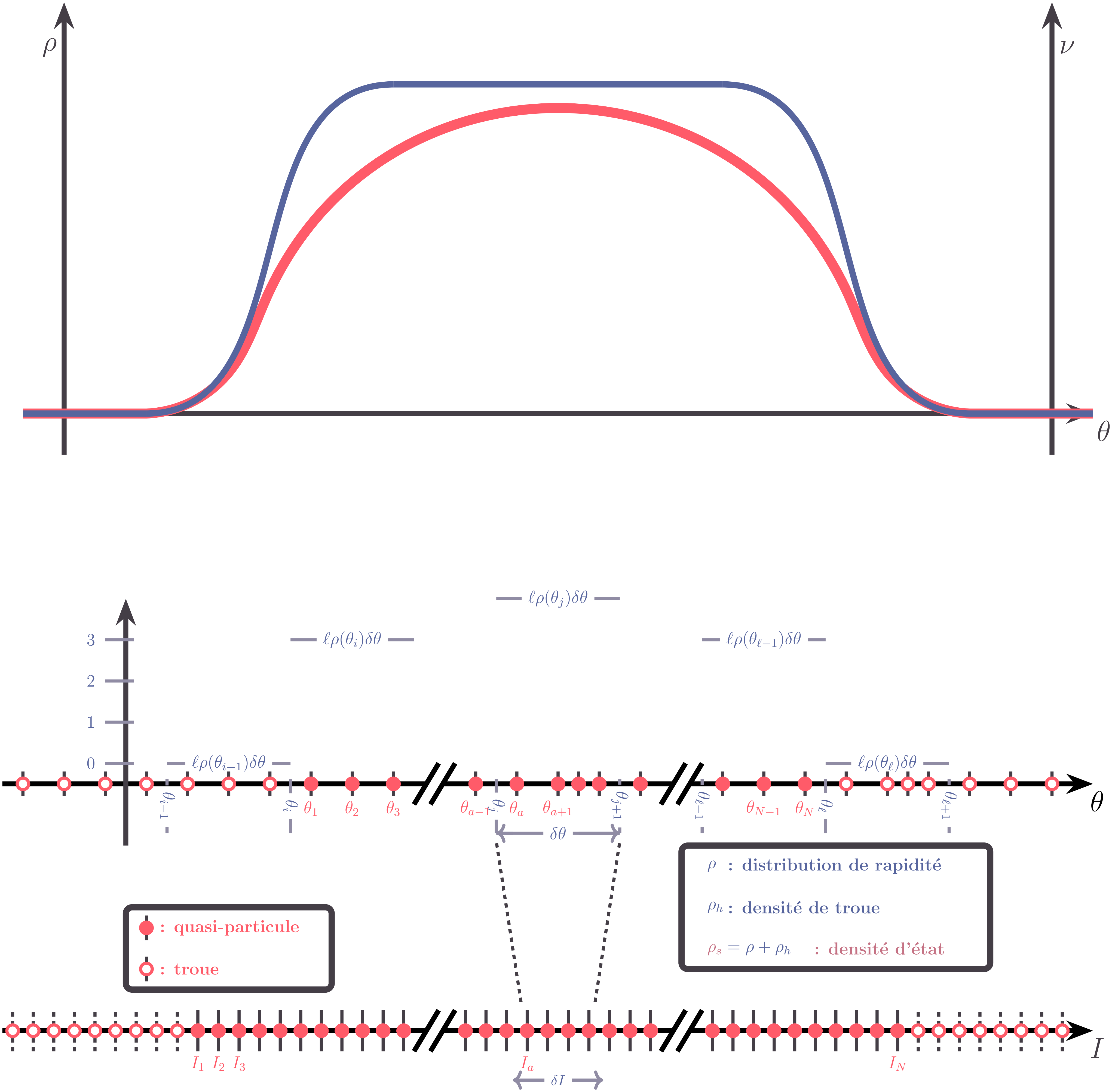
$-\frac{N-1}{2}$
$+\frac{N-1}{2}$
I



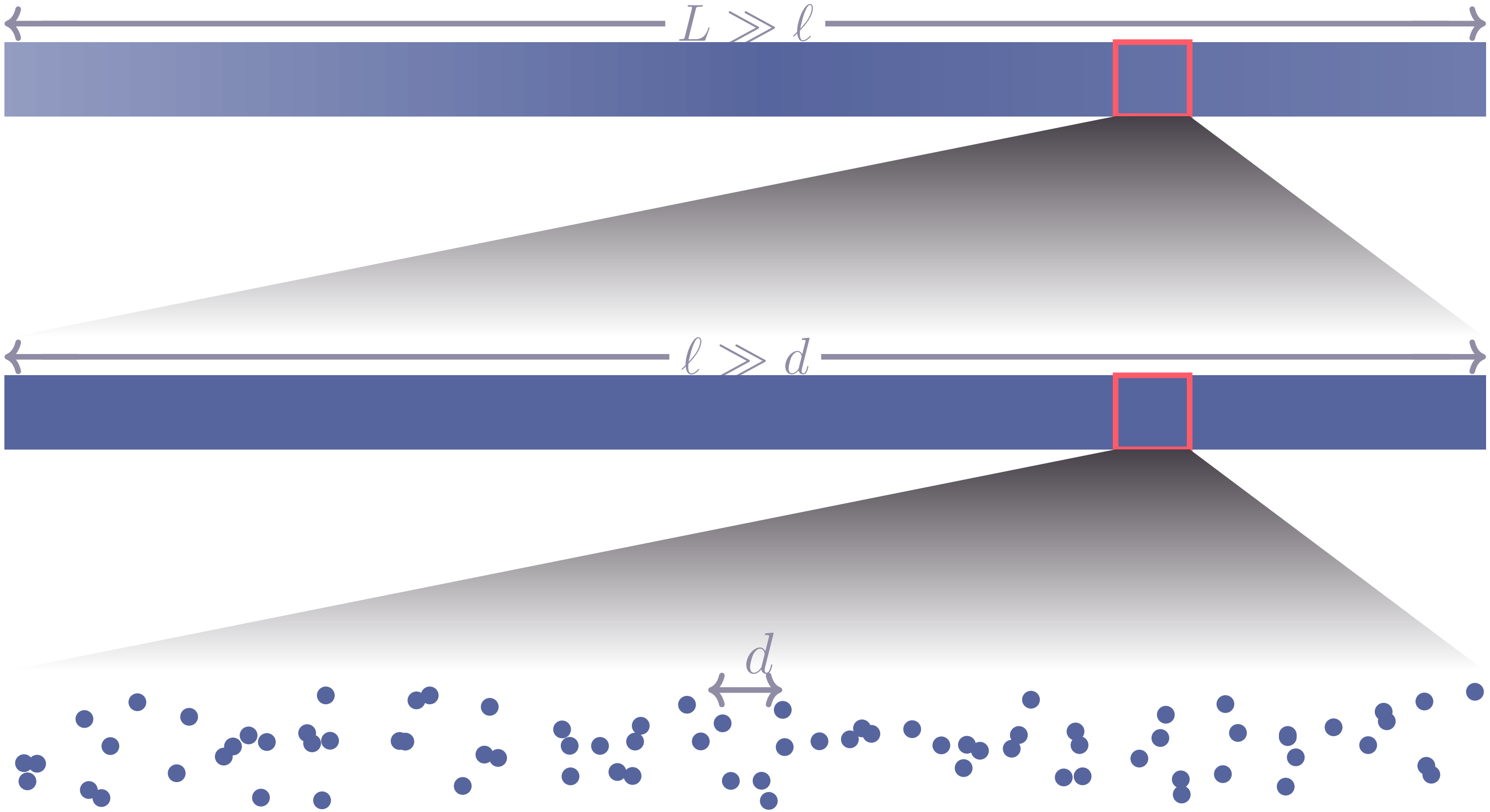
ρ
ν
θ



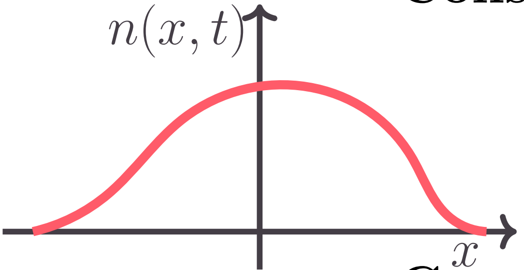
ℓρ(θ_j)δθ
ℓρ(θ_i)δθ
ℓρ(θ_{ℓ−1})δθ
ℓρ(θ_{i−1})δθ
ℓρ(θ_ℓ)δθ
3
2
1
0
θ_{i−1}
θ_i
θ_1
θ_2
θ_3
θ_{a−1}
θ_j
θ_a
θ_{a+1}
θ_{j+1}
θ_{ℓ−1}
θ_{N−1}
θ_N
θ_ℓ
θ_{ℓ+1}
θ
δθ
ρ : distribution de rapidité
ρ_h : densité de troue
ρ_s = ρ + ρ_h : densité d'état
: quasi-particule
: troue
I_1
I_2
I_3
I_a
I_N
I
δI



ρ
ν
θ
$\ell\rho(\theta_j)\delta\theta$
$\ell\rho(\theta_i)\delta\theta$
$\ell\rho(\theta_{\ell-1})\delta\theta$
$\ell\rho(\theta_{i-1})\delta\theta$
$\ell\rho(\theta_\ell)\delta\theta$
3
2
1
0
θ_{i-1} θ_i θ_1 θ_2 θ_3 θ_{a-1} θ_j θ_a θ_{a+1} θ_{j+1} $\theta_{\ell-1}$ θ_{N-1} θ_N θ_ℓ $\theta_{\ell+1}$
$\delta\theta$
ρ : distribution de rapidité
ρ_h : densité de troue
$\rho_s = \rho + \rho_h$: densité d'état
: quasi-particule
: troue
I_1 I_2 I_3 I_a I_N
I
δI



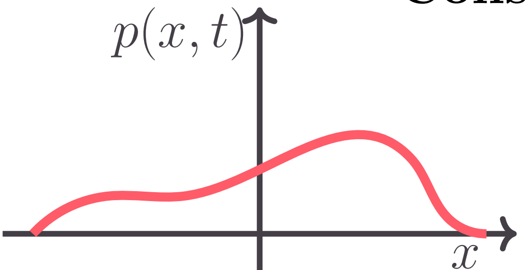
$L \gg \ell$
$\ell \gg d$
d



Conservation of the atom number :

$$\int dx\, n(x,t) = N$$

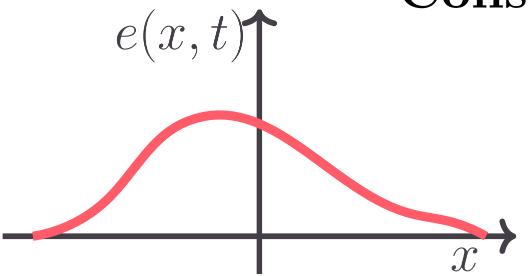
$$\partial_t n + \partial_x j_n = 0$$



Conservation of total momentum:

$$\int dx\, p(x,t) = P$$

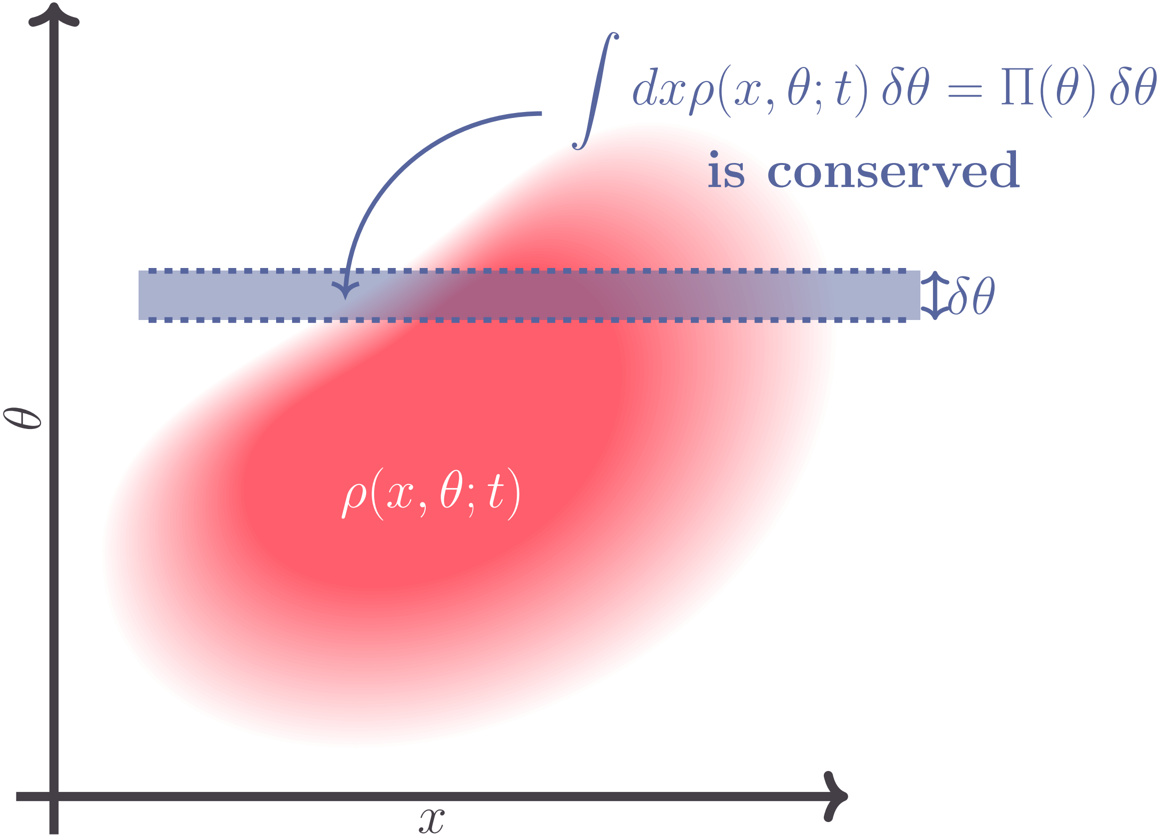
$$\partial_t p + \partial_x j_p = 0$$



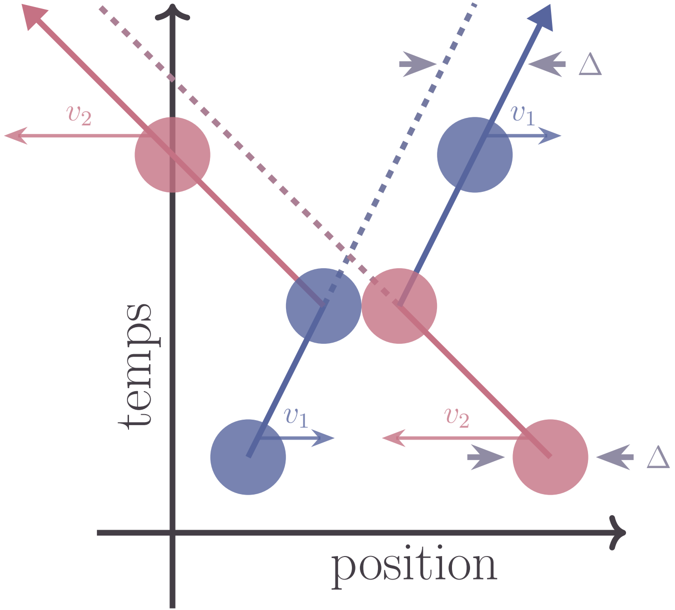
Conservation of total energy :

$$\int dx\, e(x,t) = E$$

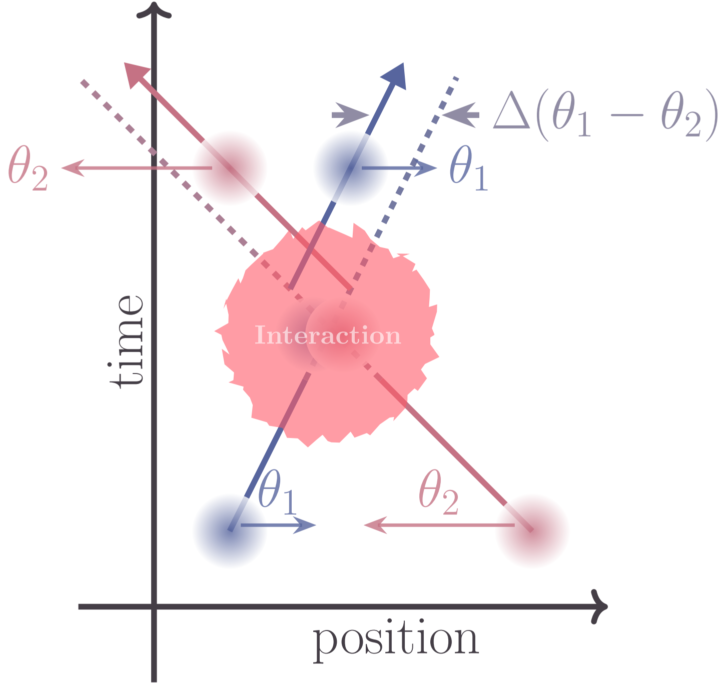
$$\partial_t e + \partial_x j_e = 0$$

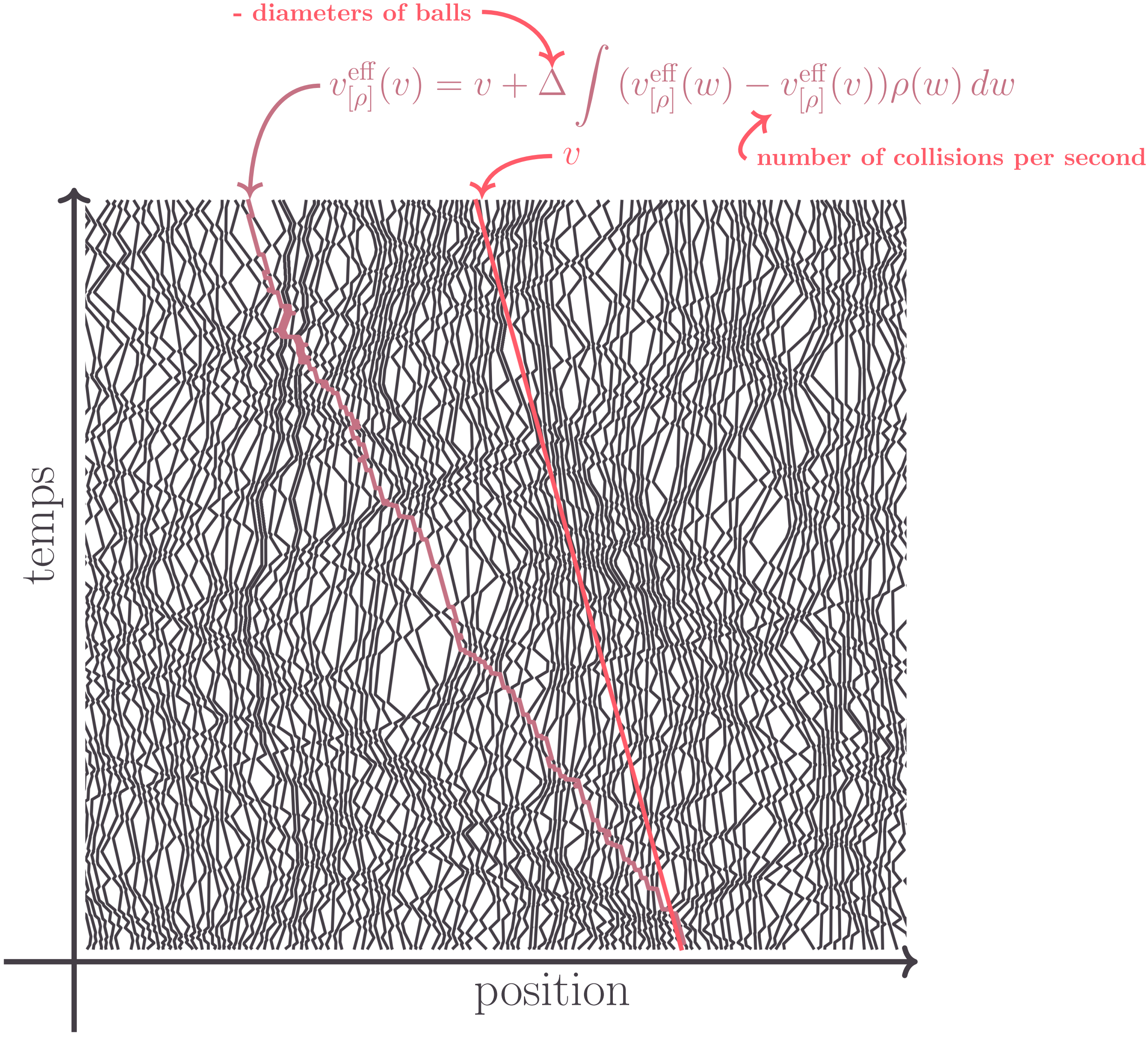


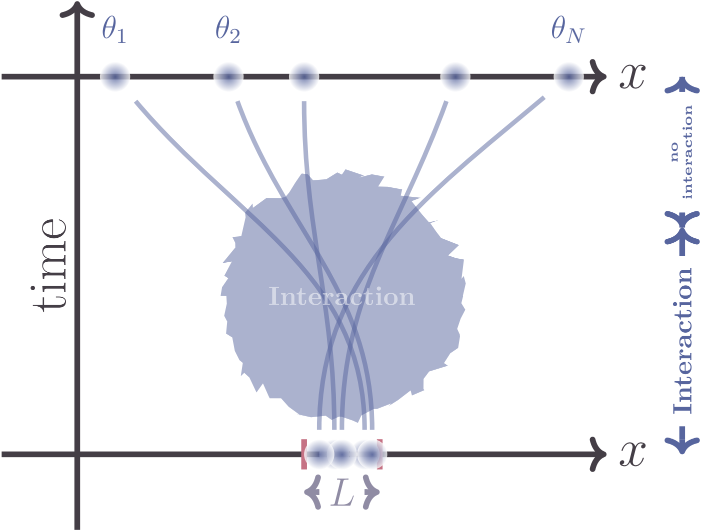
$\int dx\, \rho(x,\theta;t)\,\delta\theta = \Pi(\theta)\,\delta\theta$
is conserved
$\delta\theta$
$\rho(x,\theta;t)$
θ
x



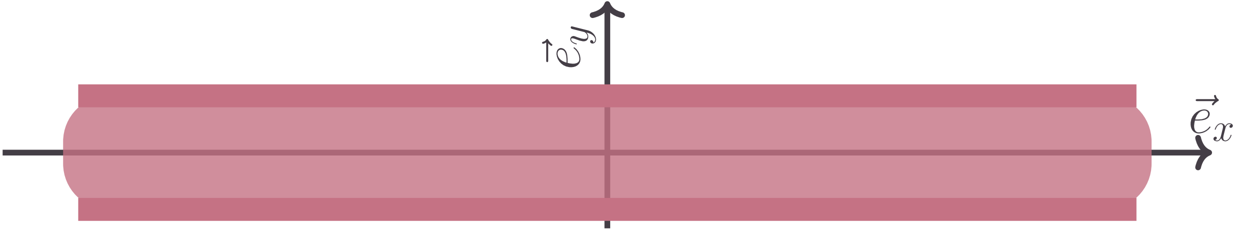
temps
position
v_1
v_2
v_1
v_2
Δ
Δ

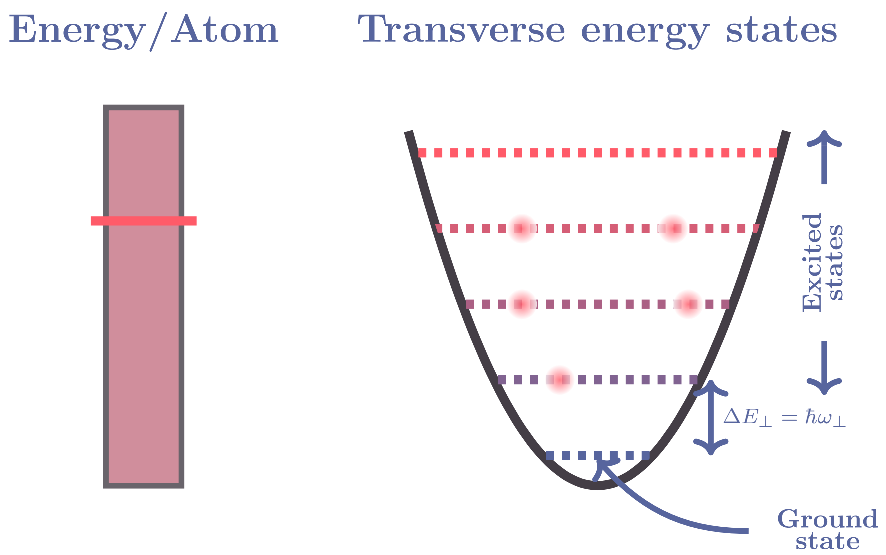

$\Delta(\theta_1 - \theta_2)$
θ_2
θ_1
time
Interaction
θ_1
θ_2
position

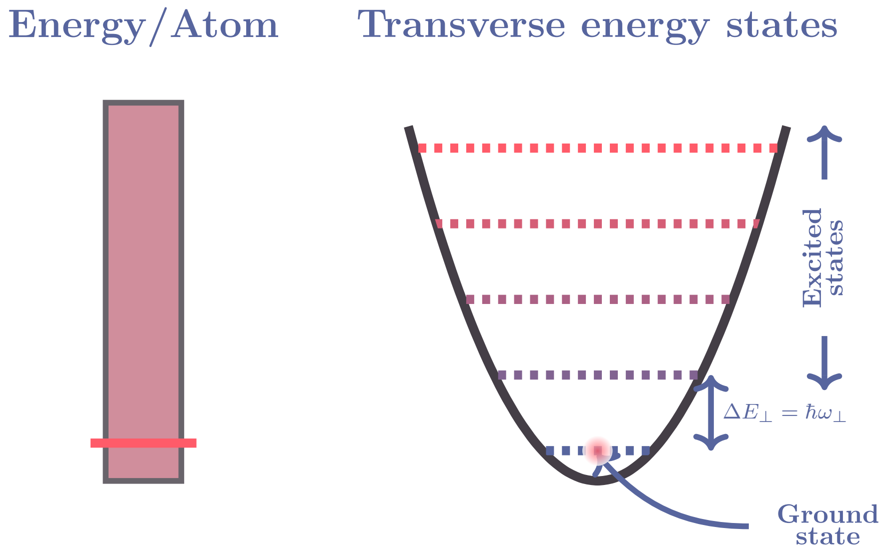

- diameters of balls
$v^{\mathrm{eff}}_{[\rho]}(v) = v + \Delta \int (v^{\mathrm{eff}}_{[\rho]}(w) - v^{\mathrm{eff}}_{[\rho]}(v))\rho(w)\, dw$
v
number of collisions per second
temps
position

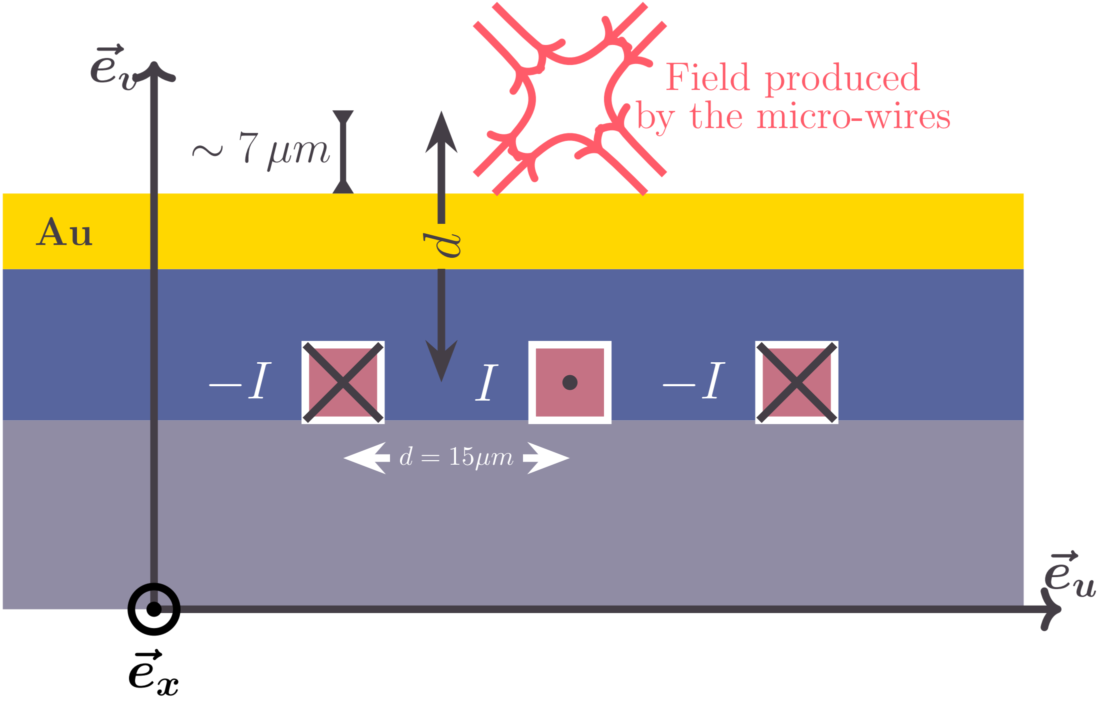


θ_1
θ_2
θ_N
x
no interaction
time
Interaction
Interaction
x
L

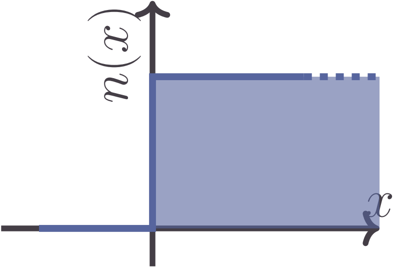

$\vec{e}_y$
$\vec{e}_x$


Energy/Atom
Transverse energy states
Excited states
$\Delta E_\perp = \hbar\omega_\perp$
Ground state

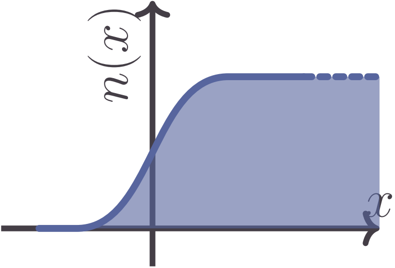

Energy/Atom
Transverse energy states
Excited states
$\Delta E_\perp = \hbar\omega_\perp$
Ground state



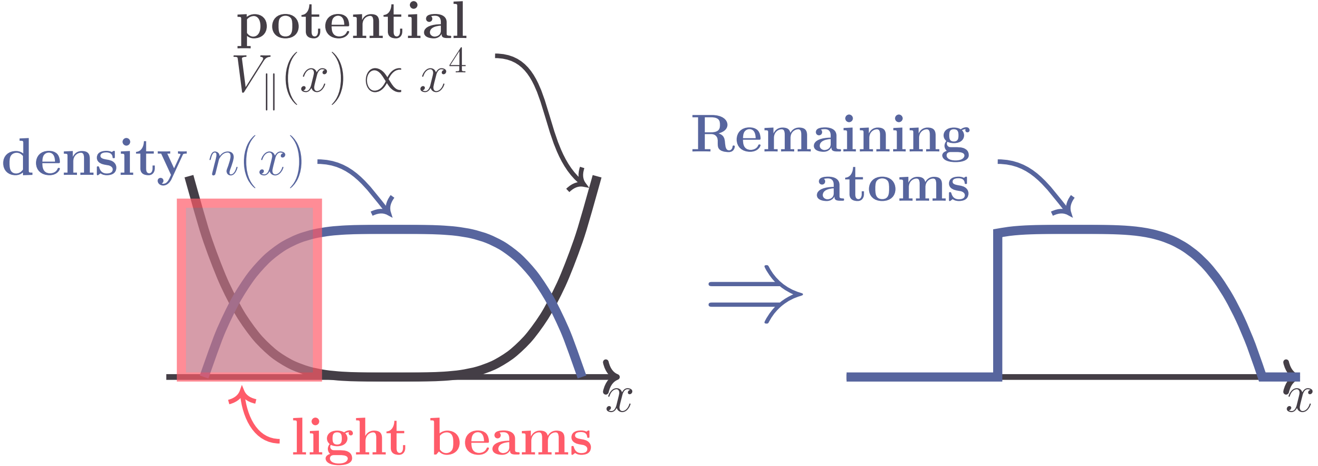
$\vec{e}_v$
$\sim 7\,\mu m$
d
Field produced
by the micro-wires
Au
$-I$
I
$-I$
$d = 15\mu m$
$\vec{e}_u$
$\vec{e}_x$



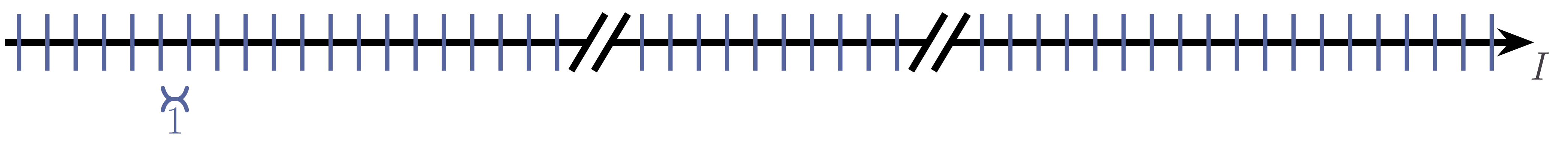
$n(x)$
x

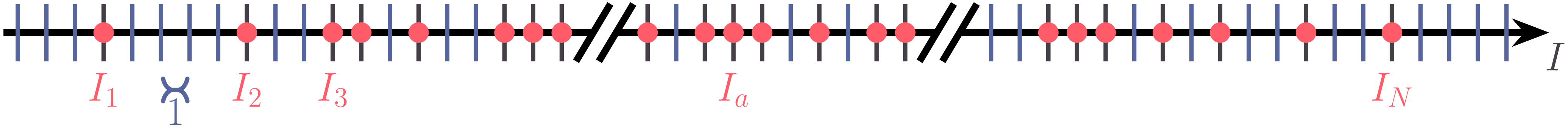


$n(x)$
x

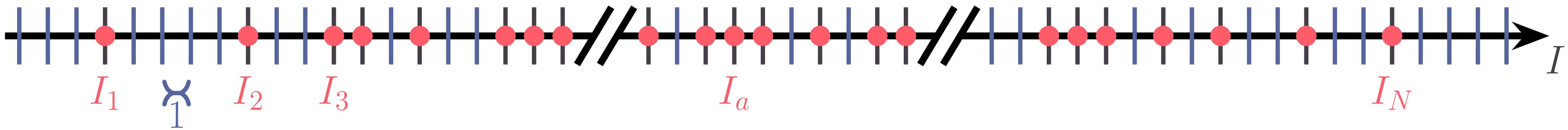


potential
$V_{\parallel}(x) \propto x^4$
density $n(x)$
light beams
x
$\Rightarrow$
Remaining atoms
x

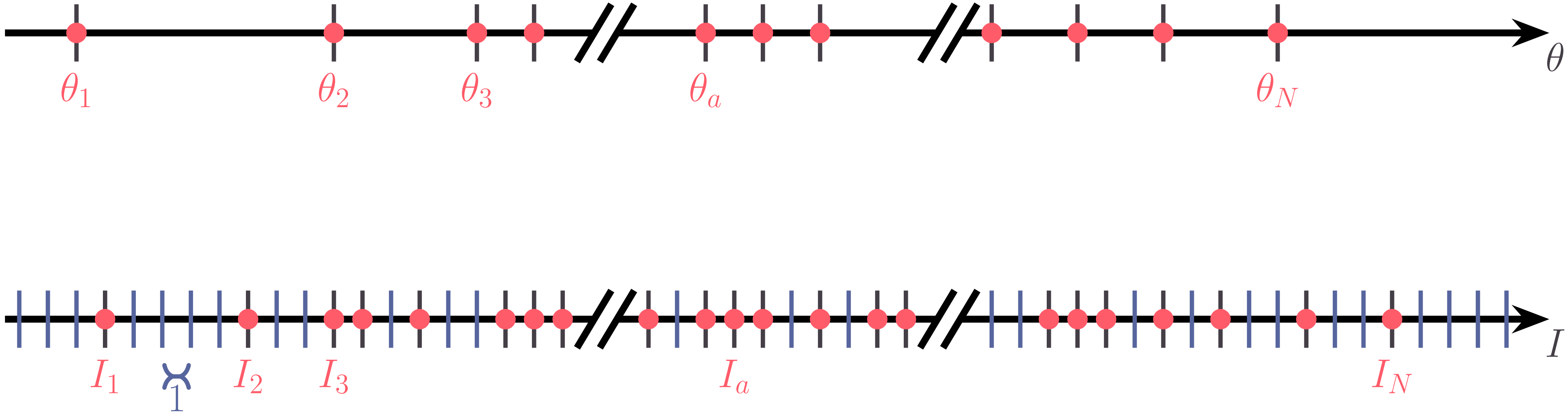

1
I



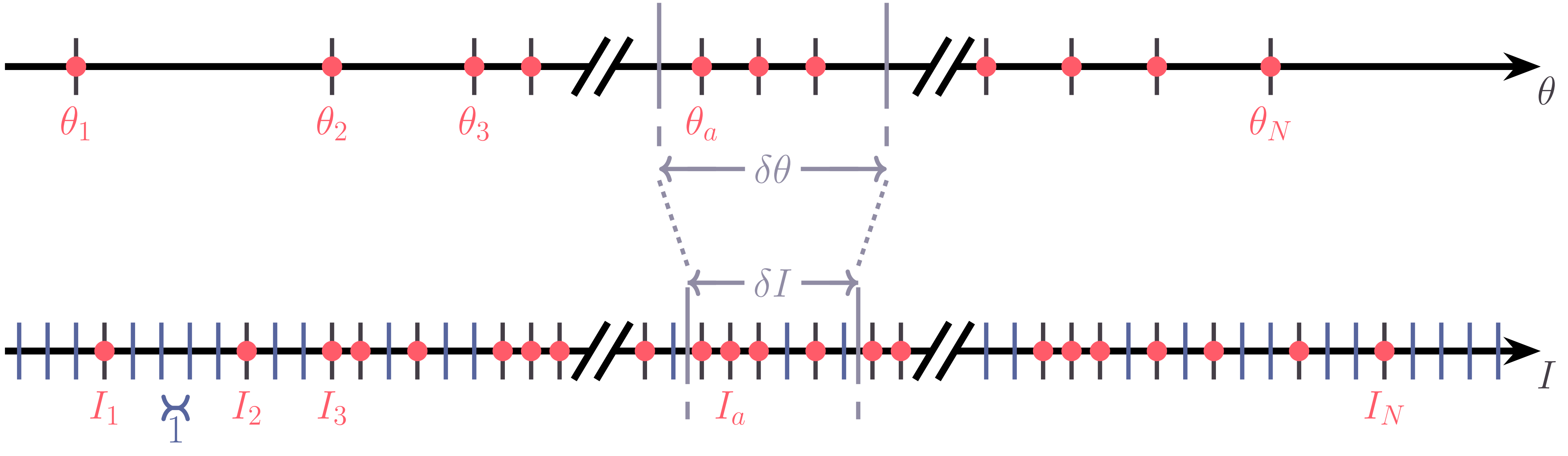
I_1
1
I_2
I_3
I_a
I_N
I



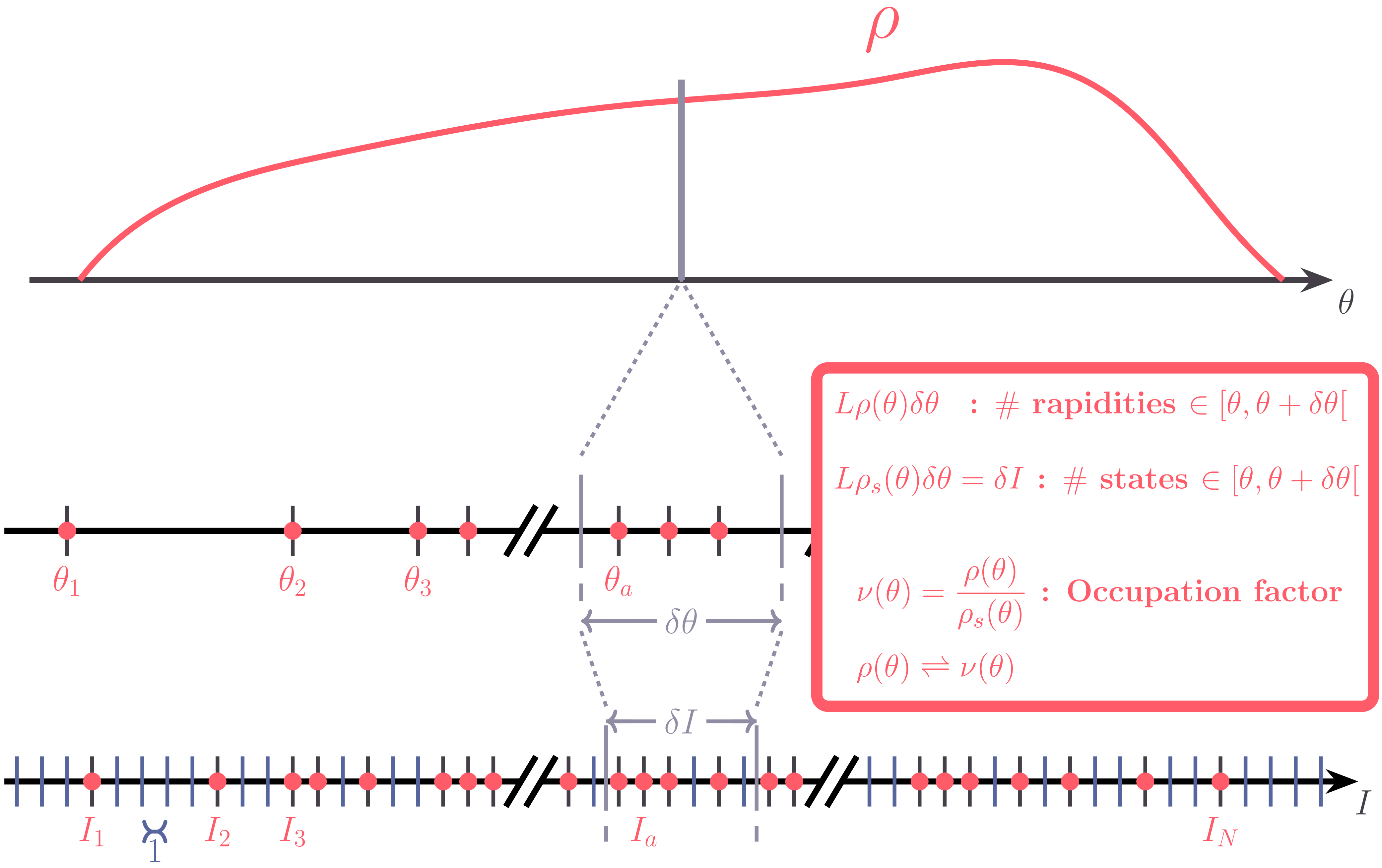
I_1
1
I_2
I_3
I_a
I_N
I



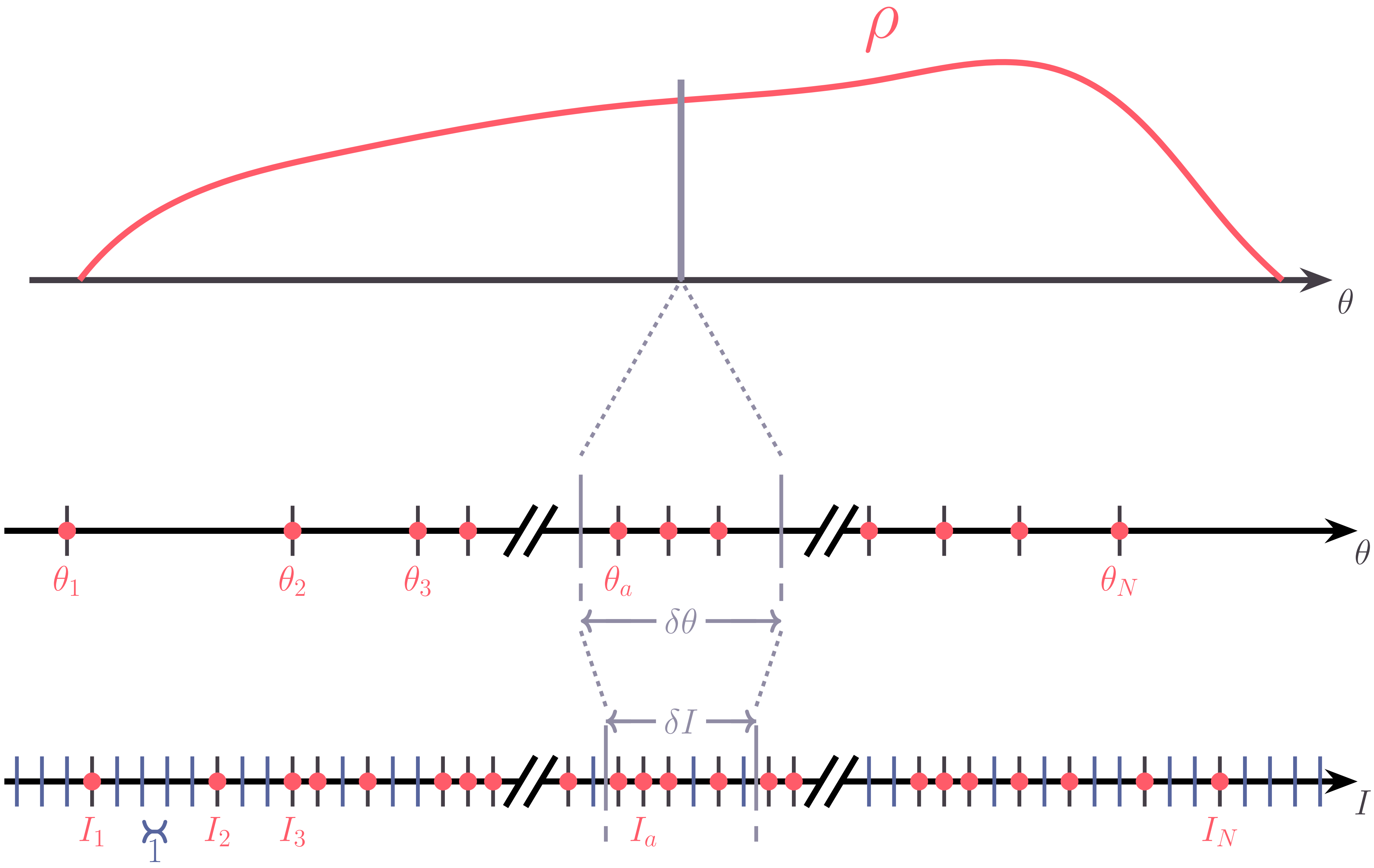
θ_1
θ_2
θ_3
θ_a
θ_N
θ
I_1
1
I_2
I_3
I_a
I_N
I



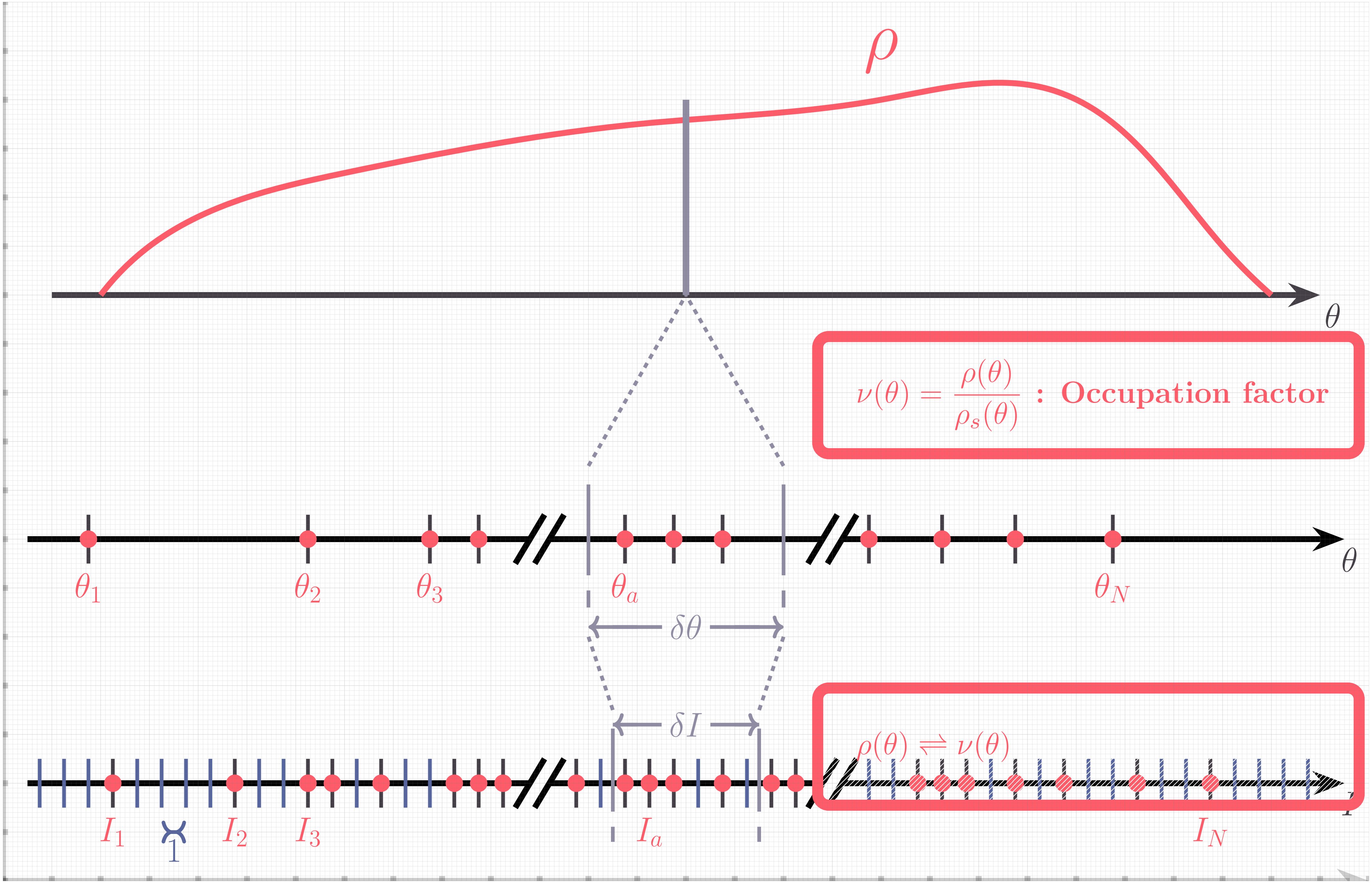
θ_1
θ_2
θ_3
θ_a
θ_N
θ
$\delta\theta$
δI
I_1
1
I_2
I_3
I_a
I_N
I



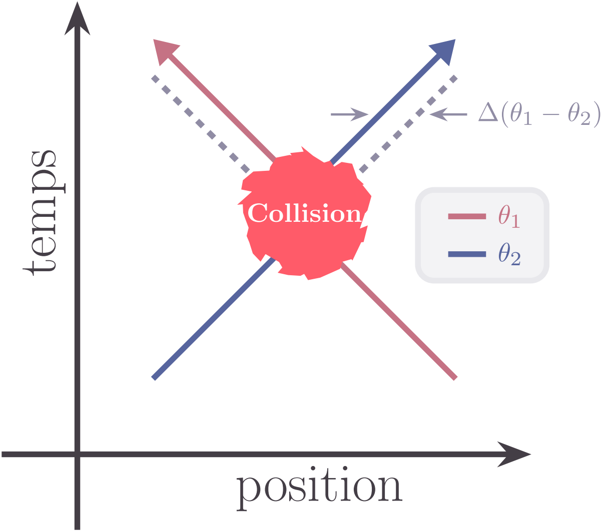
ρ
θ
$L\rho(\theta)\delta\theta$: # **rapidities** $\in [\theta, \theta+\delta\theta[$
$L\rho_s(\theta)\delta\theta = \delta I$: # **states** $\in [\theta, \theta+\delta\theta[$
$\nu(\theta) = \dfrac{\rho(\theta)}{\rho_s(\theta)}$: **Occupation factor**
$\rho(\theta) \rightleftharpoons \nu(\theta)$
θ_1
θ_2
θ_3
θ_a
$\delta\theta$
δI
I_1
1
I_2
I_3
I_a
I_N
I



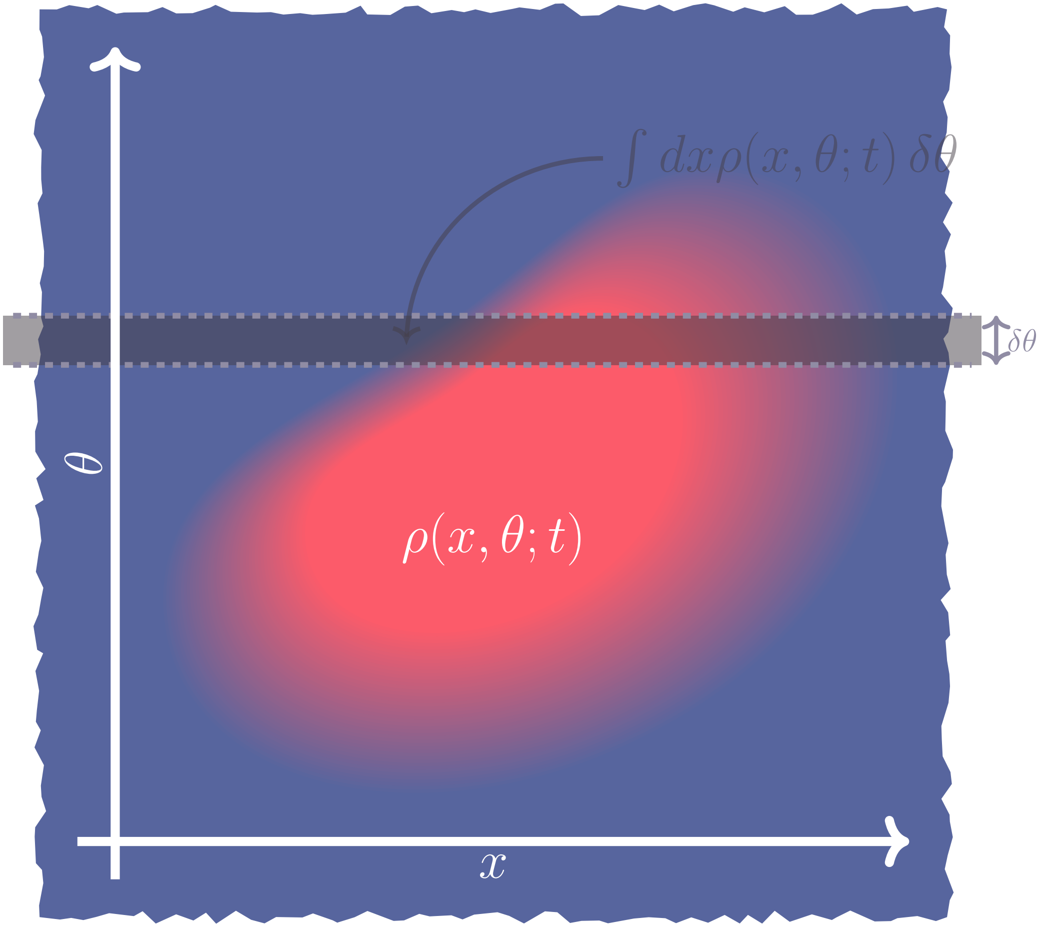
ρ
θ
θ_1
θ_2
θ_3
θ_a
θ_N
θ
$\delta\theta$
δI
I_1
I_2
I_3
I_a
I_N
I
1



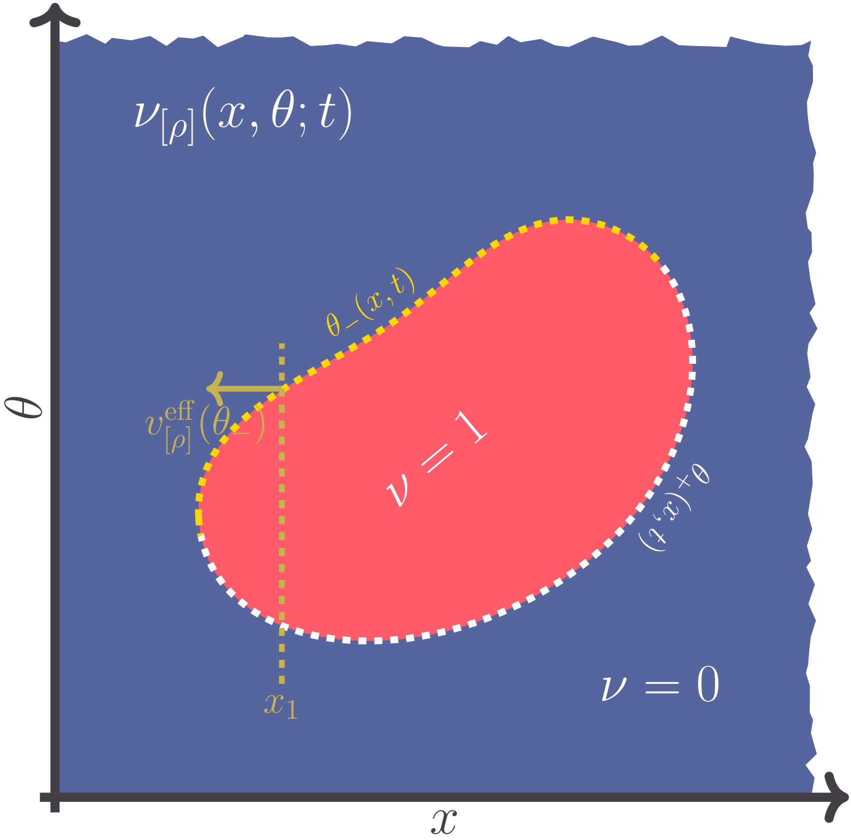
ρ
θ
$\nu(\theta) = \frac{\rho(\theta)}{\rho_s(\theta)}$: **Occupation factor**
θ
θ_1
θ_2
θ_3
θ_a
θ_N
$\delta\theta$
δI
$\rho(\theta) \rightleftharpoons \nu(\theta)$
I_1
1
I_2
I_3
I_a
I_N



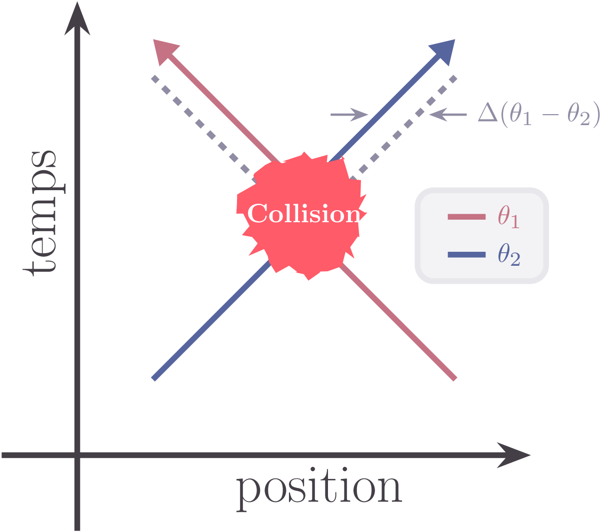
temps
position
Collision
$\Delta(\theta_1 - \theta_2)$
θ_1
θ_2



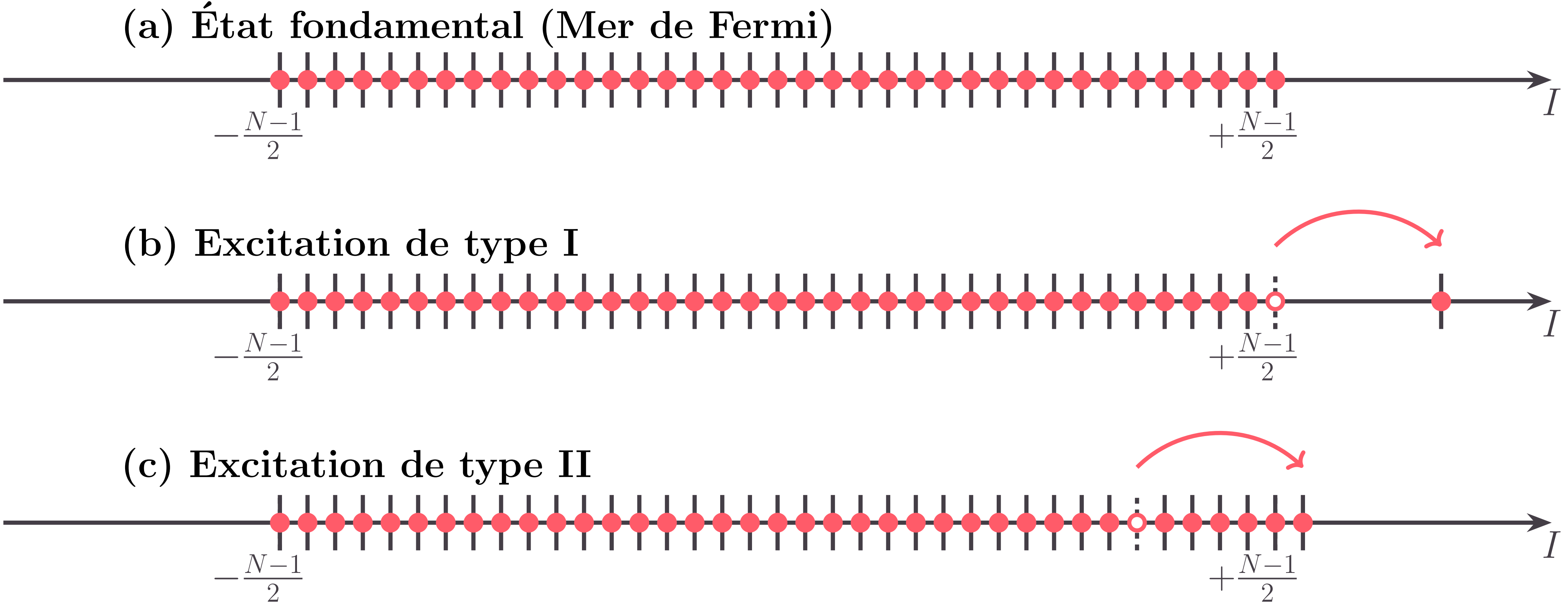
$\int dx\, \rho(x,\theta;t)\,\delta\theta$
$\delta\theta$
θ
$\rho(x,\theta;t)$
x

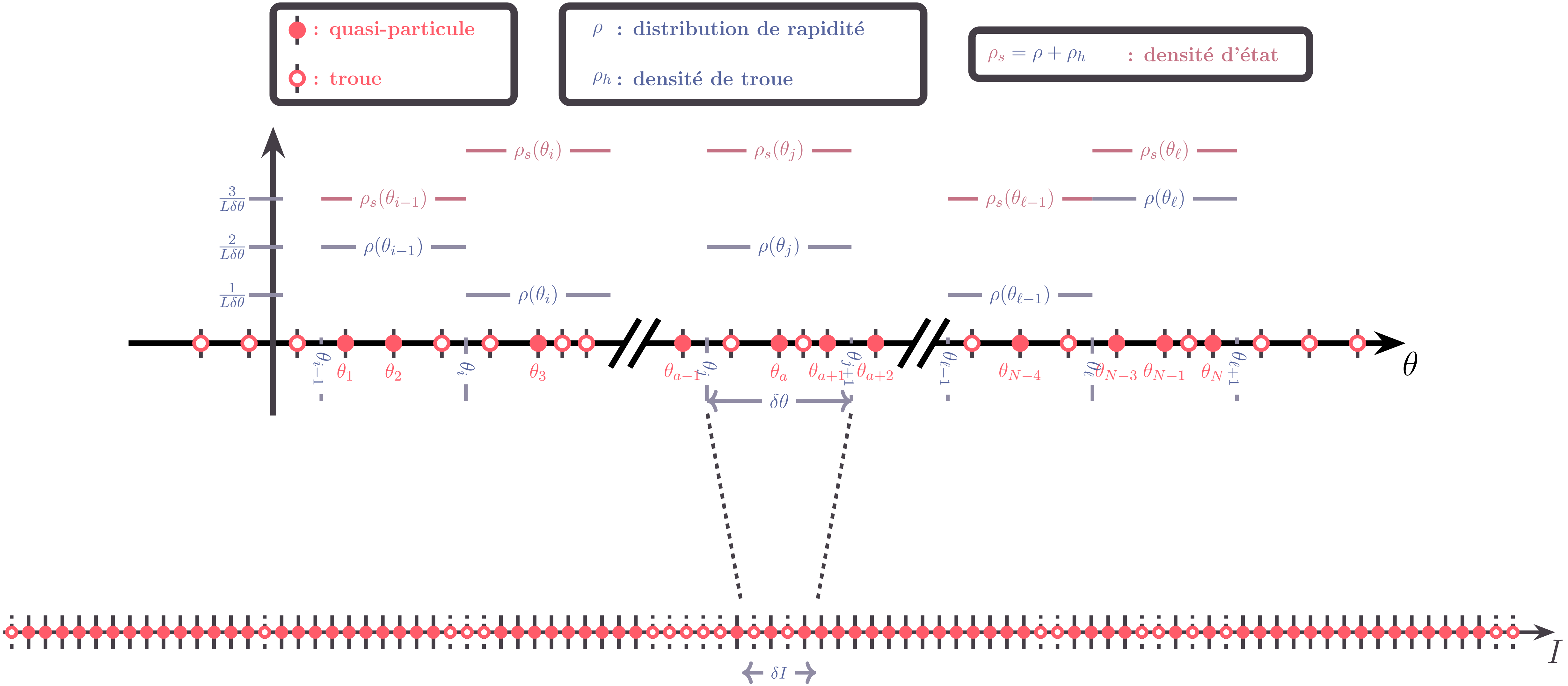


$\nu_{[\rho]}(x,\theta;t)$
$\theta_-(x,t)$
$v^{\mathrm{eff}}_{[\rho]}(\theta_-)$
$\nu = 1$
$\theta_+(x,t)$
$\nu = 0$
x_1
θ
x

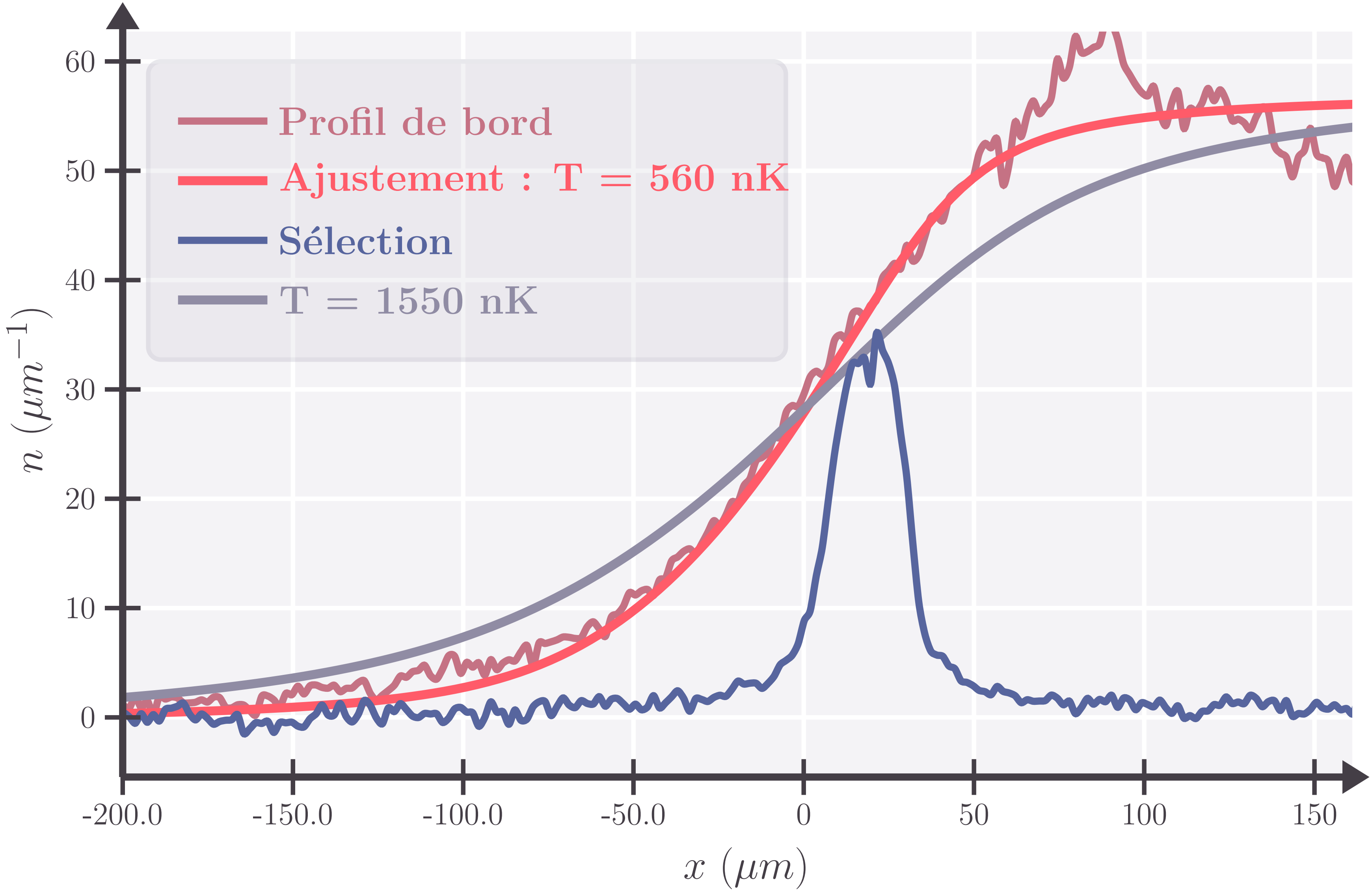


temps
position
Collision
$\Delta(\theta_1 - \theta_2)$
θ_1
θ_2

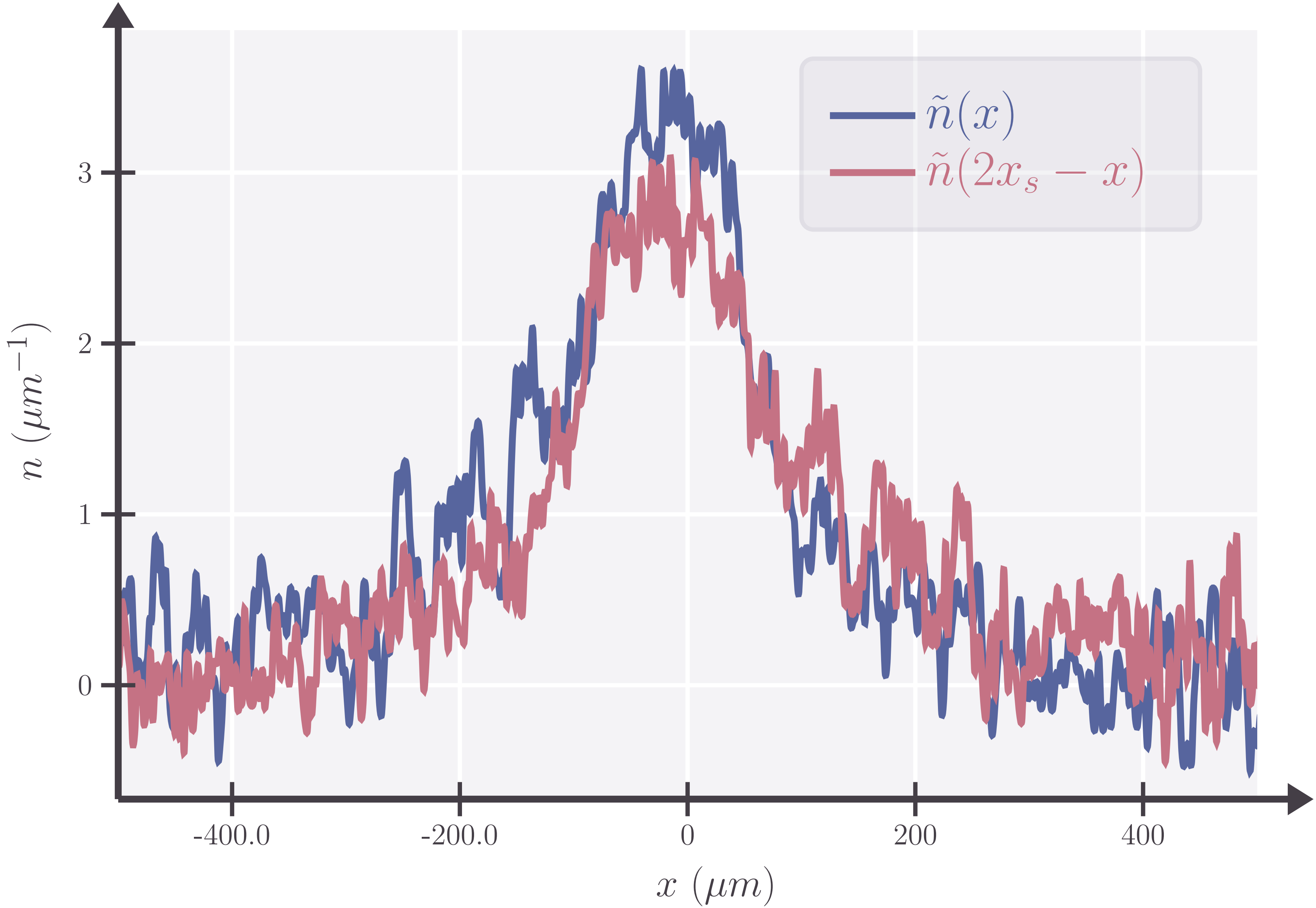

(a) État fondamental (Mer de Fermi)
(b) Excitation de type I
(c) Excitation de type II
$-\frac{N-1}{2}$
$+\frac{N-1}{2}$
I

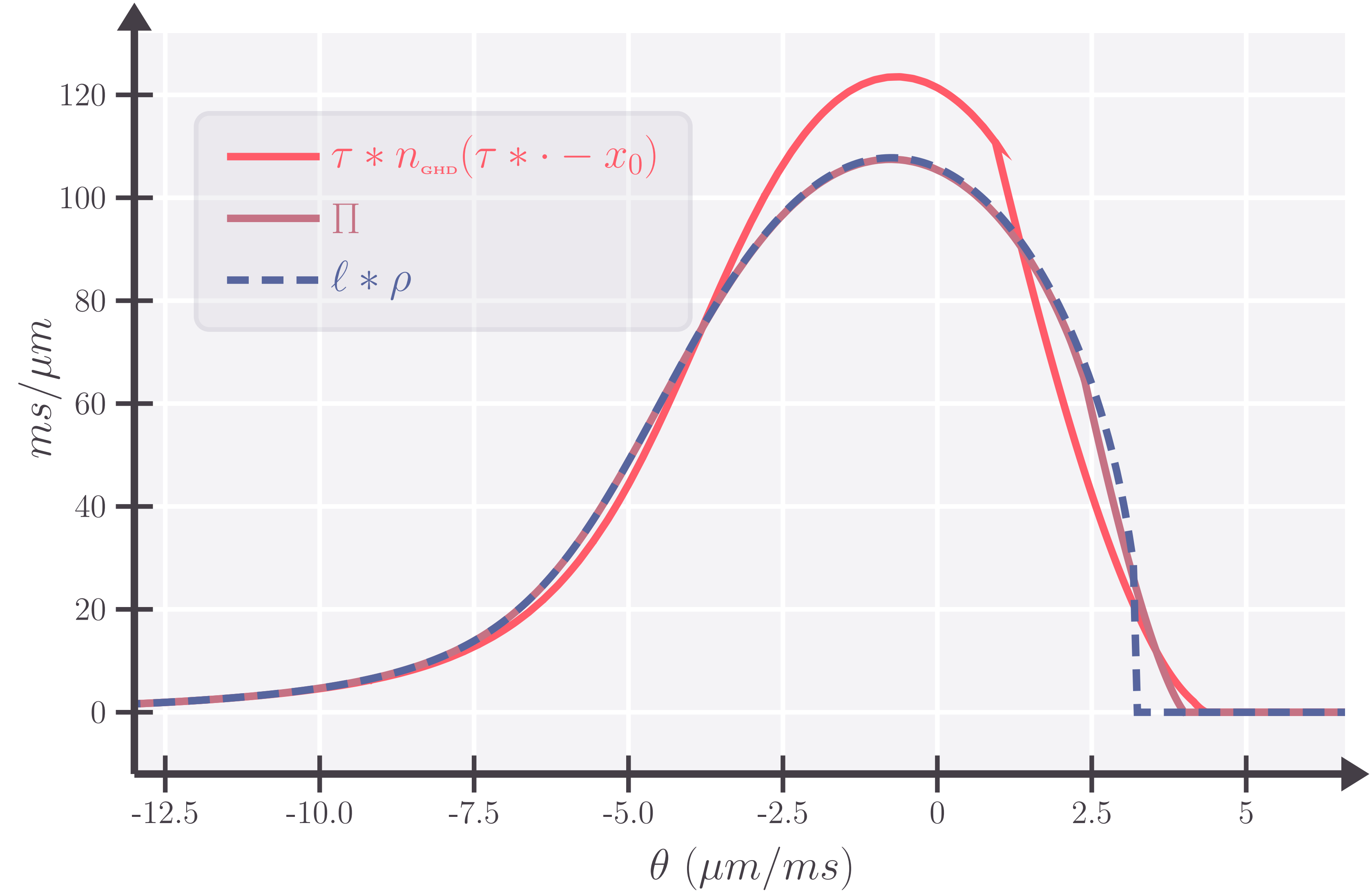


: quasi-particule
: troue
ρ : distribution de rapidité
ρ_h : densité de troue
$\rho_s = \rho + \rho_h$: densité d'état
$\rho_s(\theta_i)$
$\rho_s(\theta_j)$
$\rho_s(\theta_\ell)$
$\rho_s(\theta_{i-1})$
$\rho_s(\theta_{\ell-1})$
$\rho(\theta_\ell)$
$\rho(\theta_{i-1})$
$\rho(\theta_j)$
$\rho(\theta_i)$
$\rho(\theta_{\ell-1})$
$\frac{3}{L\delta\theta}$
$\frac{2}{L\delta\theta}$
$\frac{1}{L\delta\theta}$
θ_1
θ_2
θ_3
θ_{a-1}
θ_a
θ_{a+1}
θ_{a+2}
θ_{N-4}
θ_{N-3}
θ_{N-1}
θ_N
θ
$\delta\theta$
δI
I

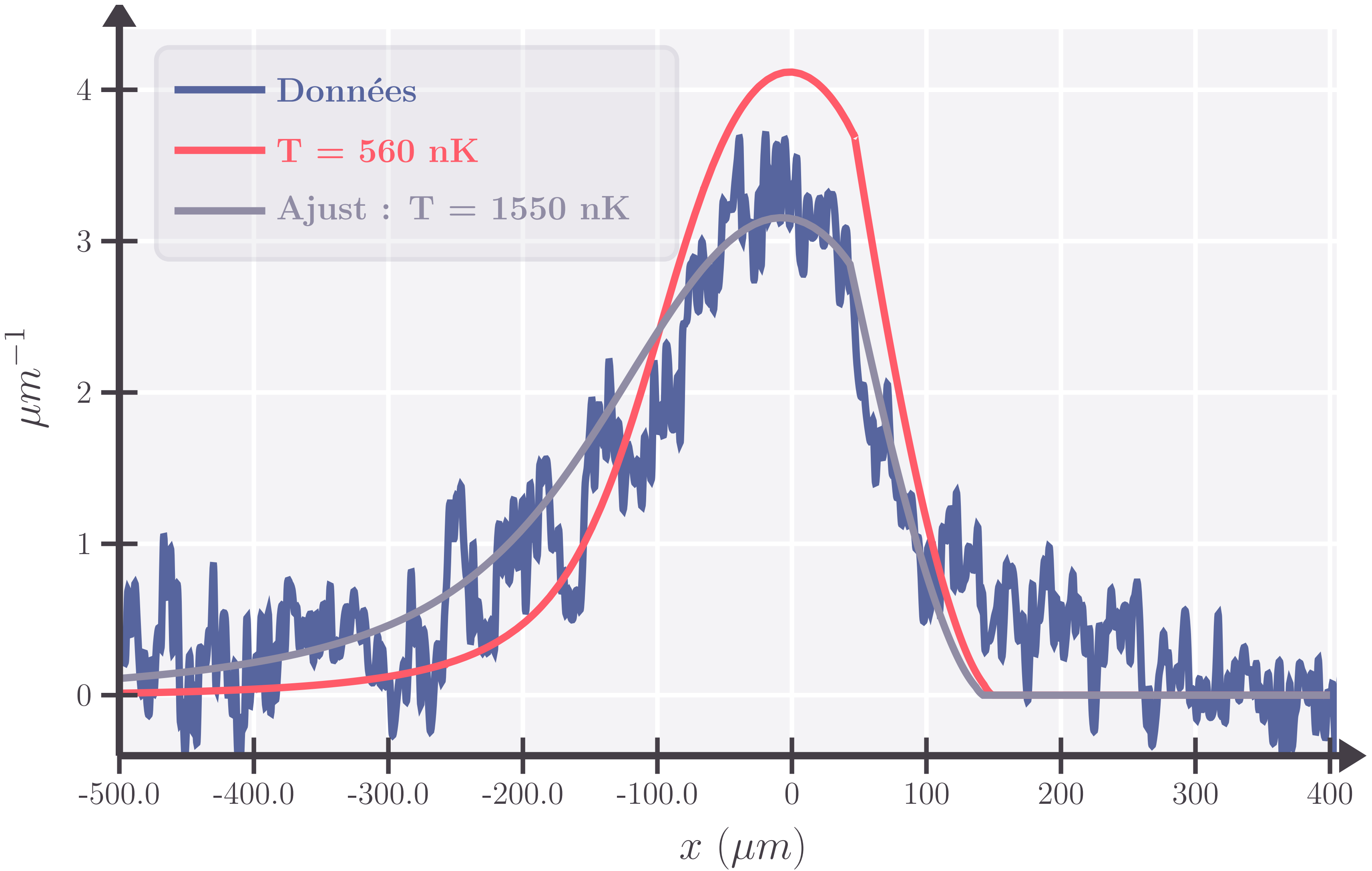


Profil de bord
Ajustement : T = 560 nK
Sélection
T = 1550 nK
60
50
40
30
20
10
0
n (μm^{-1})
-200.0
-150.0
-100.0
-50.0
0
50
100
150
x (μm)


$\tilde{n}(x)$
$\tilde{n}(2x_s - x)$
$n\ (\mu m^{-1})$
0
1
2
3
-400.0
-200.0
0
200
400
$x\ (\mu m)$



$\tau * n_{\text{GHD}}(\tau * \cdot - x_0)$
Π
$\ell * \rho$
120
100
80
60
40
20
0
$ms/\mu m$
-12.5
-10.0
-7.5
-5.0
-2.5
0
2.5
5
$\theta\ (\mu m/ms)$


Données
T = 560 nK
Ajust : T = 1550 nK
μm^{-1}
0
1
2
3
4
-500.0
-400.0
-300.0
-200.0
-100.0
0
100
200
300
400
$x\ (\mu m)$